Sec. VIII-x: the error analysis, verified on the campaign

drop-in; figure `fig_sec8_error.png`

0.1 What is guaranteed, and what is measured

Two different kinds of statement are made about the accuracy of the identification and they must not be merged. Equation (108) is a *guarantee*: given a scale, a CAD inertia and the excitation protocol, the terms the model neglects cannot displace the identified threshold by more than $\bar{\rho}$, which is 12.04 mN m for roll and 9.95 for pitch, or 0.400 and 0.322 mm of equivalent offset. It says nothing about sensor noise, rig repeatability or anything else it did not model. The repeatability reported below is a *measurement* of the whole error, with no model in it at all. Quoting either as if it were the other would misstate both.

0.2 The bound, checked forwards

The guarantee is verified as described in Sec. **??**: each perturbation is evaluated along the measured trajectory, propagated through the deviation dynamics and correlated against the weight of (107). All 140 runs fall inside, with a median displacement of 0.049 mN m, 0.161 at p90 and 0.927 at worst against the 12.04 allowed.

That margin is not an accident and it has a closed form. Writing $x = C_2\tau_{\text{end}}$, the gravity remainder behaves as $\rho_\varphi \propto \varphi^2 \sim e^{2C_2 s}$ near the window end, so $e_\omega \sim e^{2x}$ while $\bar{\rho} \sim e^{2x}/x$ and therefore the envelope of (93) grows as $E \sim e^{3x}/x$:

$$\frac{|e_\omega|}{E} \;\sim\; x\,e^{-x}. \tag{VIII.1}$$

Both grow with the window; the bound grows by one whole factor of e^x more, because the Chebyshev step of (93) puts the window average of ρ underneath a kernel that is largest exactly where ρ is smallest. Equivalently, keeping the covariance that Chebyshev discards turns the inequality into an identity,

$$\frac{|e_\omega|}{E} = 1 + \operatorname{corr}(\rho, h_\omega)\,\mathrm{CV}_\rho\,\mathrm{CV}_h, \tag{VIII.2}$$

and lengthening the window makes both coefficients of variation grow faster than the correlation weakens. Measured, the realised deviation is 3.2% of the bound at $\dot{M} = 0.10$ and 20.8% at 1.20; the reduction factors evaluated at their suprema account for a further factor of 1.5 to 2.1, roughly constant across the rates.

This says how informative the guarantee is and nothing else, and the distinction is worth making explicitly because the two are easily conflated. A larger occupancy means the bound is closer to the truth, not that the truth is larger or smaller. On the ρ channel the fast ramp is in fact slightly *worse*: the deviation is less aligned with the onset direction there (31.4° against 23.0°, so 82% of it is absorbable rather than 89%), but $\Delta M_{\text{crit}} = \dot{M}\,\delta$ and δ falls only 2.3× across the rate range while $\dot{M}$ rises twelvefold, so the channel contributes 0.35 mN m at $\dot{M} = 0.10$ and 1.82 at 1.20. The recommendation on excitation rate is made separately below, on measured grounds, and does not rest on this paragraph.

0.3 The fit against its cap

Because the nominal response is itself an admissible member of the fitted family, the minimiser cannot do worse than it does, and with the sensor noise added by the triangle inequality

$$\|\hat{r}\| = \min_g \|y - g\| \le \|y - f\| \le \|e_\omega\| + \|n\| \le \|E\| + \|n\|. \tag{VIII.3}$$

The comparison needs the residual split first. e_ω is a Duhamel integral of a smooth ρ and carries essentially nothing above a few times $1/\tau_{\text{end}}$, so residual content faster than that cannot be the small-angle or ground-effect remainder whatever else it is; charging the model for it is not a test of the model. Splitting at 5 Hz (the windows are 53 samples at 100 Hz, too short for an IIR filter), the low-frequency residual lies inside the cap for 114 of 140 runs. Including the high-frequency content the figure is 88/140.

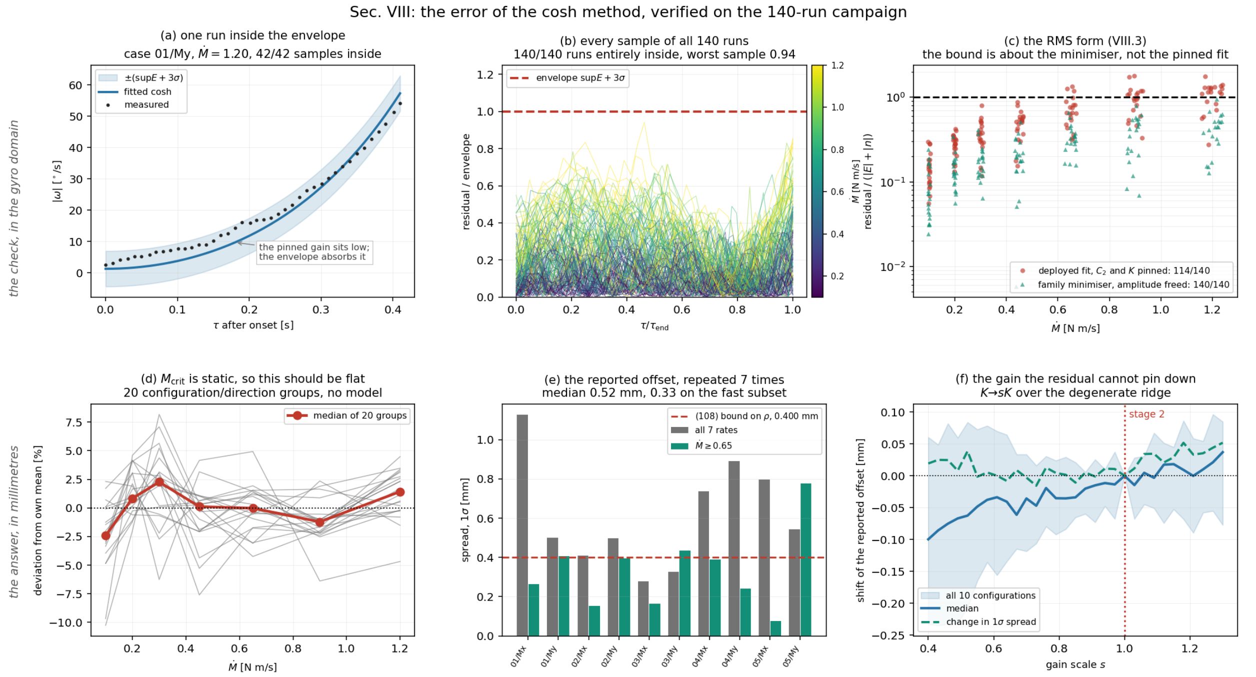

Figure 1: The error of the identification, measured on the 140-run campaign. The top row is the check and is in degrees per second throughout; the bottom row is the reported answer and is in millimetres. A per-direction residual is not an offset error, and the two are kept apart deliberately. *(a)* one run against the envelope: the measured rate, the fitted response, and $\pm(\sup E + 3\sigma)$ with E the propagated bound (93) and σ the in-window disturbance. The measured points sit systematically above the fitted curve — that is the pinned gain, discussed below. *(b)* the same test on every sample of every run, normalised so the envelope is unity: all 140 runs are entirely inside, the worst single sample reaching 0.94. *(c)* the RMS form (VIII.3). The deployed fit, with C_2 and K pinned per configuration, leaves 26 runs outside, all at the fast end; the minimiser over the same family, which is what the bound is about, leaves none. *(d)* the identified critical moment against the ramp rate for all twenty configuration/direction groups, each scaled by its own mean; M_{crit} is a static property of the rig, so the ideal is a flat line and every departure is error, with no model in the comparison. *(e)* the reported half-sum offset, repeated at seven rates per configuration, against the bound of (108). *(f)* what the pinned gain is worth: the reported offset as K is swept over the whole ridge along which the residual is degenerate, a factor of three. It moves by at most 0.10 mm in the median and the spread does not move at all.

The split also identifies what the rest is. Above 5 Hz the residual carries 1.02°/s while the pre-onset record carries 0.14, a factor of seven: before the onset the vehicle is supported by every landing gear and after it pivots on one edge with far less damping, so the pre-onset floor understates the in-window disturbance and must not be used as $\|n\|$.

The envelope, sample by sample. The RMS form averages, so it is worth stating the pointwise version as well, which is what Fig. 1(a,b) shows. Taking the envelope at its supremum over the window and widening it by three times the in-window disturbance, every sample of every run is inside: 140/140 runs, with the worst single sample reaching 0.94 of the envelope. Against the tighter, time-varying $E(\tau)$ the same test holds for 98.2% of all samples and 98/140 runs entirely. Nothing in the record leaves the band the error analysis predicts.

That is a measurement, and it has a proof, from the same two hypotheses (VIII.3) already uses. The naive pointwise statement $|r| \le |e_\omega| + k\sigma_n$ is false at the onset, where $E(\tau) \propto \sinh C_2\tau$ vanishes and the residual does not, because the fit is selected by the whole window; what it omits is the displacement of the fit itself, and that displacement is expressible in e_ω and n alone. Since ω_{nom} is a member of the family and $\hat{\omega}$ is a global minimiser over it, $\|\hat{\omega} - \omega_{\mathrm{nom}}\| \le 2\|e_\omega + n\|$ by the triangle inequality, with the factor dropping to one when C_2 is a constant and the minimiser is therefore the orthogonal projector P onto the two-dimensional span. The displacement lies in that span, where $v = Pv$ gives $|v_i| \le \sqrt{P_{ii}}\|v\|$ exactly, so

$$|r_i| \;\le\; E_i + \sqrt{P_{ii}}\,\|E\| + k\sigma_n\big(1 + \sqrt{P_{ii}}\big), \tag{VIII.4}$$

which holds at every sample of all 140 runs with a worst ratio of 0.611. The constants do not grow with the record because $\operatorname{tr} P = 2$ regardless of N: over the seven rates $\max_i \sqrt{P_{ii}}$ runs 0.35 to 0.40 and the displacement term 1.09 to 1.29 times $\sup E$, giving $\max|r| \lesssim 2.3 \sup E + 1.4\, k\sigma_n$. Two terms are added to the right-hand side and neither is covered by $\bar{\rho}$: the deployed fit's pinned amplitude contributes $|\Delta C_1| u(\tau)$, which is (VIII.3)'s 26 runs seen pointwise, and a calibration error $C_2 = C_2^*(1 + \varepsilon_2)$ moves the nominal out of the subspace and contributes $\|(I - P)\omega_{\mathrm{nom}}\|_\infty$, worth 1 to 4% of $\sup E$ per 2% of ε_2. The derivation is given in full separately.

The twenty-six that fail. They are not scattered. Every one is at $\dot{M} \ge 0.65$ N m/s (3, 9 and 14 of 20 at the three fastest rates, none at the four slowest), they come from eight of the ten configurations, and the worst misses by 1.80 with a median exceedance of 1.15 at the fastest rate. A shortfall that respects the rate and ignores the configuration is not a bad calibration of one vehicle; it is something in the inequality itself, and (VIII.3) has exactly one hypothesis that the campaign does not satisfy.

The first equality there requires $\hat{g}$ to be the minimiser over the model family. The deployed estimator's is not: C_2 and K are pinned once per configuration by the stage-one fit and frozen for every run of that configuration, so the per-run fit is a *constrained* minimiser and can only leave a larger residual than the free one the bound is about. Freeing the amplitude alone, with C_2 held at its physical value $\sqrt{W z_{CoM}/J_P}$ and the onset held where the estimator put it, brings all 140 runs inside — not 114, and not 139, but every one, at every rate.

The freed amplitude sits 11.3% below the pinned $K\dot{M}$ (p10 22.1%, p90 0.1%), and that gap accounts for the excess in size and not merely in sign: an amplitude error ΔC_1 leaves $\Delta C_1(\cosh C_2\tau - 1)$ behind, and band-limiting that shape the same way predicts the measured excess to within 0 to 13% at every rate, a median of 3%. Only a small part of the 11% is the ramp itself rounding off near the allocation cap, which lowers the post-onset slope by 1 to 2.6%; the remainder is the stage-one gain, and it is the same 5 to 12.5% contamination that the calibration stage is independently shown to inherit from e_ω in its own fitting data. Its residual grows as $\dot{M}$ while the cap falls with the window, which is why the failures appear only at the fast end.

Two consequences follow, and they point in opposite directions, so both are stated. The check as posed is a test of the *deployed* fit, not of ρ: the model family provably contains a curve inside the cap for every run, so the small-angle and ground-effect analysis is not what fails. But the pinned gain is a real bias, and correcting it moves the per-direction threshold by 7.2 mN m in the median and 24.3 at the fastest ramp, which is 0.81 mm of equivalent offset. What protects the reported answer is the two-sided difference of the next subsection: the gain is common to both directions while $\dot{M}$ is not, so the shift is odd and the half-sum moves by 1.30 mN m, 0.043 mm — a factor of 5.5 smaller than per direction and an order of magnitude below the 0.327 mm the campaign repeats to. The in-window disturbance accounts for the remaining margin: estimating its low-frequency part from a quiet window of equal length, split at the same cutoff, and carrying the measured in-window content above 5 Hz down through that ratio raises 114 to 124 on its own.

Why the gain is not simply re-fitted. The obvious response is to take the amplitude the data prefers and re-pin K on it. It cannot be done, and the reason is the degeneracy the calibration stage is built around: with the onset swept over the window, a smaller amplitude with an earlier onset trades against a larger amplitude with a later one, so the residual is flat along a ridge in (C_2, K). Scaling the frozen K by s with C_2 held and reprocessing all 140 runs at each s shows the ridge directly. The residual has no interior minimum: it falls monotonically from $s = 1$ to $s \approx 0.43$, and eight of the ten configurations place their optimum at the low edge of whatever grid is offered. A 57% smaller K asserts $W z_{CoM}$ is 2.3 times what the scale reads. The count inside the cap follows the residual — 114/140 at $s = 1$, 139/140 at $s = 0.43$ — so it is not a criterion for setting K either; it is bought by making the gain physically wrong. Stage two therefore selects K by the ramp-invariance of M_{crit} instead, and that criterion does have an interior optimum, at $s = 0.97$ in the median with the per-configuration values scattered 0.76 to 1.21 on fourteen runs each. That is scatter about unity, not a bias away from it.

Reprocessing at the correction it does license changes nothing that is reported: the offset moves 0.019 mm in the median and 0.068 at worst, and the half-sum spread goes 0.522 to 0.533 mm (with each configuration on its own optimum, 0.022 mm and 0.522 to 0.506 mm). The insensitivity is the point. Across the whole ridge — K scaled by a factor of three, 0.40 to 1.30 — the reported offset moves at most 0.10 mm in the median and the spread stays within 0.507 to 0.574 mm, because the gain is common to both tip directions while $\dot{M}$ is not and the half-sum removes it. A constant the residual cannot pin down at all is one the answer does not depend on.

0.4 Repeatability, with no model in it

The campaign supplies a model-free accuracy measurement, because M_{crit} is a *static* property of the rig — the moment at which the vehicle is on the verge of tipping over a landing-gear edge — and cannot depend on how fast the moment was ramped to reach it. Identifying it at seven rates per configuration and per direction therefore repeats a measurement whose true value is known to be constant, and Fig. 1(d) is that test.

	per direction	reported half-sum
scatter σ_M	21.3 mN m	—
as equivalent offset	0.71 mm	—
1σ across the rates	—	0.28 to 1.13 mm
median	—	0.522 mm
median, $\dot{M} \geq 0.65$	—	0.327 mm

The two columns are consistent with each other without adjustment: two directions with independent moment-domain errors give a half-sum spread of $\sigma_M/\sqrt{2} = 0.50$ mm, and the measured median is 0.522.

Which domain. A mislocated onset is an error in *time*, so it would show a flat σ_t and a σ_M rising with $\dot{M}$; a vehicle that does not tip at exactly the same moment twice is an error in *moment*, so σ_M is flat and σ_t falls as $1/\dot{M}$. Measured, σ_M runs 13.7 to 33.5 mN m with a coefficient of variation of 34% while σ_t runs 11.4 to 225 ms at 97%: the error is in the moment.

The alternative can be excluded quantitatively rather than by inspection. If the scatter were a within-run disturbance seen through the onset, then for a disturbance of fixed amplitude $\sigma(\Delta M_{\text{crit}}) \propto \sqrt{C_2/B(x)}$, so the short window — the fast ramp — would scatter *more*, by a factor 5.72 across the rate range. It scatters 0.70, an eightfold disagreement in the opposite direction. The scatter is a run-to-run change in where the vehicle actually tips: landing-gear settling, floor friction and repositioning between runs.

The slowest ramp. One condition departs systematically. Pooling the identified M_{crit} less each configuration's own mean, $\dot{M} = 0.10$ N m/s reads -26.70 mN m against $+4.45$ for the rest (Welch $t = -4.34$, $p = 0.0002$), with 75% of those runs below their own mean. As an onset error that is 267 ms, matching the 225 ms of σ_t measured there, and its sign says the onset is found *early* — what a slow rise emerging from noise does to a residual minimiser. It is not ρ, which contributes 0.3 to 1.8 mN m on these channels, twenty times too small and with the opposite rate trend. Restricting the protocol to $\dot{M} \geq 0.65$ N m/s removes it and takes the reported spread from 0.522 to 0.327 mm.

0.5 What the two-sided difference removes

The offset is read from the half-sum $M_{\text{off}} = (M_+ + M_-)/2$, and the two runs tip over different edges with restoring arms $A_\pm = l_p \mp a$. In mirrored coordinates both obey the same equation, so in signed φ the arm term of the deviation forcing carries $\text{sign}(\dot{M})$ while the $Wz\sin\varphi$ term does not: the arm term is antisymmetric and cancels. Simulating the full nonlinear tip-over with a 2 mm offset and fitting with the deployed estimator, 98.70% of the per-direction bias cancels at every rate, leaving about 1 μm. The ground-effect channel cancels as well: β_M flips sign with the pivot edge — it is proportional to $\sum_i (l_{y,i} + l_p)^2 b^\dagger_{i,\varphi}$, and the allocation basis reverses — so the bilinear term of (79) is odd and contributes 0.6 to 0.9 μm instead of the 46 to 78 it would if it were even. The per-configuration calibration cancels too, being common to both directions by construction: $\pm 10\%$ of amplitude error and $\pm 5\%$ of exponent error are rejected to 99.77–99.98% and move the offset by 0.000 mm.

What does not cancel is the weight, because (34) reads the *difference* of the two critical moments, where the same bias adds. Per 1% of calibration error that is 0.027% of W from the amplitude channel and 0.077% from the exponent, the exponent being the more damaging by about three; the closed forms are $A(x)/B(x)$ and $D(x)/B(x)$ times $\dot{M}/C_2$.

0.6 Choosing the excitation rate

The protocol's seven rates permit the choice to be made on measurement rather than argument. Taking each configuration's half-sum at a given rate against its own mean over all rates isolates the error contributed by that rate:

$\dot{M}$ N m/s	post-onset samples	window s	RMS error mm	p90 mm
0.10	74	0.740	0.810	1.035
0.20	64	0.635	0.569	0.975
0.30	58	0.575	0.708	0.994
0.45	54	0.545	0.707	1.374
0.65	48	0.480	0.528	0.924
0.90	44	0.445	0.443	0.696
1.20	41	0.410	**0.422**	**0.637**

There is no interior optimum: the error falls to the fastest rate tested. That is worth checking against the obvious objection, which is that a faster ramp shortens the window and leaves fewer samples to fit. It does — 74 down to 41 — but the onset is located by $\|\chi\|^2 = C_1^2 C_2 B(x)$ with $C_1 = \dot{M}/Wz_{CoM}$, so the amplitude rises twelvefold while the sample count falls only 1.8×, and the product improves: $\|\chi\|$ grows 1.35× and the onset uncertainty σ_δ falls by 26%. Forty-one samples remain ample for a two-parameter fit.

Two effects would eventually create an interior optimum and neither has taken over by 1.20 N m/s. The onset is quantised to the sample grid, contributing $T_s\dot{M}/\sqrt{12}$ per direction, or 0.081 mm to the half-sum at the fastest rate against the 0.422 measured — under 4% of the variance. It is removable outright, by taking the vertex of a parabola through the three costs bracketing the grid minimum rather than the minimum itself; on a synthetic onset placed between samples that takes the error from 5.00 to 0.26 ms. Applied to the campaign it changes nothing measurable, the per-rate spread moving by −3.1 to +3.2%, which is what a 4% variance contribution and ten configurations predict. The refinement is retained because it removes the term analytically and because it becomes binding if the rate is raised or the logging rate lowered, not because it improves these numbers. And the pseudo-inverse allocation saturates at $M_{x,\max} = 2.371$ and $M_{y,\max} = 2.737$ N m (Figs. 13–14), which caps the usable rate from above. Within the range tested the recommendation is simply the top of it, and restricting to $\dot{M} \geq 0.65$ N m/s takes the reported spread from 0.522 to 0.327 mm.

0.7 Summary

term	how known	equivalent offset
small-angle + GE remainder	bounded a priori, (108)	≤ 0.400 mm
realised, p90	forward check, 140/140	0.005 mm
onset location	not resolved; below the next line	—
rig repeatability, per direction	measured	0.71 mm
reported offset, all rates	measured, 7× per config	**0.522** mm
reported offset, $\dot{M} \geq 0.65$	measured	**0.327** mm

The modelling approximations are therefore a bounded minority of an error the rig dominates: whatever limits this method, it is not the neglected $\sin\varphi$. That is worth stating plainly because it also justifies the change of onset model. The quadratic growth law used previously carried a *systematic* error of 12 to 31% of the window in the located onset — 22 to 58 mN m over the observed x range, comparable to or larger than the entire error measured here, and a bias rather than a scatter, so averaging does not reduce it. Retaining $Wz_{CoM}\varphi$, which is exactly what turns the quadratic into the hyperbolic cosine with $C_2^2 = Wz_{CoM}/J_P$, moves the algorithm's contribution from dominant to negligible.